

Help for command ReportTieP

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1 Description of ReportTieP

This command generate report on quality of orientation relative to a set of tie points, by default it generates these analysis by image using multiple tie-points, but it can optionnally generate result for tie points by pair. It generate result similar to **ReportGCP**, see also this command for detailed interpretation of files.

It generate three kind of information :

- statistical information in **csv** format , for individual images and camera, and possibly for specific pairs;
- **2D** visual information, in **jpg/tiff** format of the residuals, that can be exported as dense images of field of vectors;
- **3D** visual information, in **ply** format to see the global spatialisation of the residuals.

This variety of export is required because the potential problem in orientation can have many different origin : problem of internal calibration, problem due to one or several images, problem due to a specific pair of image , problem due to some part of the 3d scene.... Generally, all these export are not generated simultaneously, the user can generate a first report without optionnal parameter and then focuss and specific reports.

2 Run with default parameters

An example of run With the three defaultt parameters example on Ramses dataset:

```
MMVII ReportTieP IMG_03.*JPG V1Dense Adjust
```

It will generate statistic of residual of residual in folder `MMVII-PhgrProj/Reports/ReportTieP/Adjust_V1` it will contain :

- a file `Glob.csv` with statistic on all images;

- a file `ByCam.csv` with statistic on each individual camera ; in the current case where there is only one camera these two files will be identical;
- a file `ByImage.csv` with statistic on each individual image.

The statistic contains residual for given percentile that can be modified with optional parameter `Perc`. It will also generate a dense image for the used cameras in the folder `MMVII-PhgrProj/VISU/ReportTieP/Adjust_V1Dens` if the optional parameter `DoImResCalib` is set to `true` (its default value).

3 Optional parameters for per/image visualisation

If we are interested by specific pair of images, we must use tie-point that comes from initial pairs and not merge tie points. So :

- statistic on residual for all pairs of images will be computed, they will be saved in the file `ByPairs.csv` of the report folder; they can be modified with optional parameter `Perc`;
- parameter `PatImFV` , is a pattern that is used as filter for selecting the image for which we generate the residual as a field of $2D$ vector superposed to each image.
- parameter `PatImRes` , is a pattern that is used as filter for selecting pairs of image for which we generate dense visualisation;

An example of use with Ramses dataset :

```
MMVII ReportTieP *.JPG V1Dense Adjust PatImRes="IMG_035.*.JPG" PatImFV="IMG_035[3-6].JPG"
```

Will generate 10 images of dense residual and 4 images of vector field. The vector field images is controlled by parameter `ParamFV` , see `ReportGCP`.

4 Optional parameters for per/pair statistic and visualisation

If we are interested by specific pair of images, we must use tie-point that comes from initial pairs and not merged tie points. So :

- we must use parameter `InTieP` to activate computation by pairs and specify the folder containing the tie points of pairs pairs;
- static for all pairs will be generated in file `ByPairs.csv`, using parameter `Perc`;
- the optional parameter `PatImRes` specify the images for which we generate dense visualization;
- the optional parameter `PairPatFV` specify the images for which we generate the pair for which we generate visualisation of residuals as vector's field;

An example of use with Ramses dataset :

```
MMVII ReportTieP IMG_03.*JPG V1Dense Adjust InTieP=V1Dense PairPatRes=IMG_035[0-3].JPG P
```

will generate file `ByPairs.csv` with 213 lines corresponding to all pairs having non empty set of tie points, one image vector fields for (`IMG_0351.JPG`, `IMG_0352.JPG`) and 6 image dense for pairs of $\{50 - 51, 50 - 52, \dots, 52 - 53\}$.

5 Optional parameters for 3d visualisation

The 3d visualisation can be used to analyse potential correlation between localisation and residual. For this the parameter optional `ParamPly` can be used :

- if it is used, a file `RTP.ply` will be generated ;
- the 3d point will be computed by bundles intersection using tie points and orientations;
- the color of each point will depend on the residual .

The parameter `ParamPly` contains 2 values, a boolean indicating if generate a coloured or a gray file, and an exponent E . For computing the colour :

- first we compute a global histogram H of residuals;
- then for each point, the residual r is converted to rank Rk with $Rk \in [0, 1]$;
- then Rk is converted to a quality Q using the formula $Q = 1 - Rk^E$ (some kind of *gamma-correction*);
- in gray mode, the colour is Q (points with high residual are black)
- in colour mode, Q is used to compute an abscissa, on the path *Red - Orange - Green* (points with high residual are red).

A consequence of this way of proceeding is that due to histogram preprocessing, if we multiply all the residuals by 2 or 10 or $\dots$, we will obtain exactly the same file. This visualization is not made for quantitative evaluation, but for qualitative localisation of possible areas with problems.